

Python Conditional Statements and Loops Assessment

Instructions

Answer all questions. Use Python programming language. Write efficient and readable code. Include comments wherever necessary.

1. Positive, Negative, or Zero

Write a program that accepts an integer from the user and determines whether it is positive, negative, or zero.

2. Even or Odd

Write a program that checks whether a given integer is even or odd.

3. Greatest of Three Numbers

Accept three numbers from the user and print the largest among them using conditional statements.

4. Leap Year Checker

Write a program to determine whether a given year is a leap year according to the Gregorian calendar rules.

5. Simple Calculator

Create a menu-driven calculator that performs addition, subtraction, multiplication, and division.

6. Print Numbers Using For Loop

Print numbers from 1 to 20 using a for loop.

7. Print Numbers Using While Loop

Print numbers from 20 to 1 using a while loop.

8. Sum of First N Numbers

Accept N from the user and calculate the sum of the first N natural numbers.

9. Count Even Numbers

Count and display the number of even integers between 1 and 100.

10. FizzBuzz

Print numbers from 1 to 50. Replace multiples of 3 with Fizz, multiples of 5 with Buzz, and multiples of both with FizzBuzz.

11. Multiplication Table

Generate the multiplication table of a given number up to 10.

12. Factorial Calculation

Calculate the factorial of a given number using loops.

13. Check Prime Number

Determine whether a given number is prime.

14. Next Prime Number

Given an integer N, find and print the next prime number greater than N.

15. Fibonacci Series

Print the first N terms of the Fibonacci sequence.

16. Find All Divisors

Print all divisors of a given integer.

17. GCD of Two Numbers

Find the Greatest Common Divisor (GCD) of two numbers without using built-in functions.

18. LCM of Two Numbers

Find the Least Common Multiple (LCM) of two numbers.

19. Count Digits

Count the total number of digits in a given integer.

20. Reverse a Number

Reverse the digits of a given integer.

21. Square Pattern

Using nested loops, print a square pattern of stars of size $N \times N$.

22. Right-Angled Triangle

Print a right-angled triangle pattern using stars.

23. Inverted Right Triangle

Print an inverted right-angled triangle pattern using stars.

24. Number Triangle

Print a number triangle where each row contains increasing numbers.

25. Floyd's Triangle

Generate Floyd's Triangle for N rows.

26. Pyramid Pattern

Print a centered pyramid pattern using stars.

27. Armstrong Number

Check whether a given number is an Armstrong number.

28. Palindrome Number

Check whether a number remains the same when its digits are reversed.

29. Strong Number

Check whether a number is a Strong Number (sum of factorials of digits equals the number).

30. Rightmost Non-Zero Digit

Find the rightmost non-zero digit of a given integer.

31. Diamond Pattern

Print a diamond-shaped star pattern using nested loops.

32. Prime Numbers in Range

Display all prime numbers between two user-specified limits.

33. Sum of Digits

Calculate the sum of all digits in a given number.

34. Decimal to Binary Converter

Convert a decimal number into binary without using built-in conversion functions.

35. Number Guessing Game

Create a guessing game where the user repeatedly guesses a secret number until correct.

36. Pattern Combination Challenge

Print a pattern where each row contains repeated occurrences of the row number.

37. Hollow Square Pattern

Print a hollow square pattern using stars.

38. Pascal's Triangle

Generate Pascal's Triangle for N rows.

39. Perfect Number Checker

Determine whether a number is a Perfect Number.

40. Mini ATM Simulation

Develop a menu-driven ATM simulation supporting balance inquiry, deposit, withdrawal, and exit.

Evaluation Scheme

Easy Questions (1-10): 20 Marks

Medium Questions (11-20): 50 Marks

Pattern Questions (21-26): 30 Marks

Advanced Questions (27-35): 90 Marks

Bonus Questions (36-40): 75 Marks

Total Marks: 265